

PII Category Toggle Test Harness

Implementation Plan — PromptMask Chrome Extension

Context

PromptMask is a Chrome extension that redacts PII using a local WebGPU-based LLM. Users configure which of **7 PII categories** to redact via toggles. The goal: only selected categories' PIIs are redacted; unselected ones are left untouched.

A known bug exists: credit card numbers are tagged **[PAN 1]** instead of **[CREDIT_CARD 1]** — cross-contamination between **governmentLegal** and **financialPayment**. The existing deterministic guard (**findDisallowedPlaceholderKeys**) can only catch this when the wrong category is *disabled*. When both are enabled, wrong labels slip through undetected.

The existing test suite covers prompt generation and the 128 toggle bitmask combos, but lacks:

- "One OFF" isolation tests (all ON except one)
 - Negative assertions (disabled PII literally unchanged in output)
 - Cross-contamination detection (wrong label, right category)
 - LLM-level eval that verifies the model actually respects toggle config
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Deliverables

1. cross_contamination.test.mjs

New file — deterministic, no model required

Pure-function tests against **findDisallowedPlaceholderKeys** for confusable PII pairs.

Test cases (~12):

- PAN vs CREDIT_CARD: both ON (documents the known gap — returns [] even with wrong label)
- PAN vs CREDIT_CARD: only financialPayment ON — catches [PAN 1]
- PAN vs CREDIT_CARD: only governmentLegal ON — catches [CREDIT_CARD 1]
- SSN vs BANK_ACCOUNT cross-contamination
- MRN vs CASE_ID cross-contamination
- GST vs INVOICE_ID cross-contamination
- DEVICE_ID vs PAN (both alphanumeric)
- Multiple simultaneous wrong labels
- buildCorrectionPrompt generates correct retry text for disallowed keys

Reuses: makeCategoryConfig pattern from redaction_category_toggle.test.mjs, imports from extension/offscreen_utils.mjs.

2. category_isolation.test.mjs

New file — deterministic, no model required

"One OFF" tests: for each of the 7 categories, disable only that one and verify:

- **buildPrompt()** excludes the disabled category's placeholder definitions
- **buildPrompt()** includes all 6 other categories' placeholder definitions
- **findDisallowedPlaceholderKeys()** catches ALL placeholder keys from the disabled category (not just the representative first one, as the existing 128-combo test does)
- **findDisallowedPlaceholderKeys()** allows all other categories' placeholders

7 iterations x 4 assertions = 28 assertions.

3. test_cross_contamination.txt

New fixture file — 16 snippets targeting confusable PII pairs

Uses the same ===== TEST SNIPPET N: Title ===== delimiter format. Each snippet adds Expected labels: metadata mapping PII values to their correct placeholder keys.

#	Scenario	Enabled Categories	Key Assertion
1	PAN + credit card	govLegal + financial	Each gets correct label
2	PAN + credit card	financial only	PAN untouched, CC redacted
3	PAN + credit card	govLegal only	PAN redacted, CC untouched
4	SSN + bank account	govLegal + financial	Each gets correct label
5	SSN + bank account	govLegal only	SSN redacted, bank untouched
6	MRN + CASE_ID	medical + business	Each gets correct label
7	MRN + CASE_ID	medical only	MRN redacted, ticket untouched
8	MRN + CASE_ID	business only	MRN untouched, ticket redacted
9	Name + username	identityContact	Correct labels within same category
10	API_KEY + ACCESS_TOKEN	credentials	Correct labels within same category
11	IP + email	network + identity	Each gets correct label
12	All 7 categories	all ON	Every PII gets correct label
13	All 7 categories	identity only	Only identity PIIs redacted
14	PAN + CC + SSN	govLegal + financial	Three-way disambiguation
15	GST + INVOICE_ID	govLegal + business	Alphanumeric disambiguation
16	DEVICE_ID + PAN	network + govLegal	Hex vs alphanumeric

4. Extend test_redaction_category_toggle.txt

7 new "one OFF" snippets (snippets 15-21)

Each contains PII from all 7 categories but disables exactly one. The assertion is that only the disabled category's PII survive untouched.

Example fixture header:

```
CATEGORY TOGGLE TEST SNIPPET 15: ALL ON except identityContact
Disabled category: identityContact
Expected untouched: NAME, PHONE, EMAIL, USERNAME, ADDRESS, DATE_OF_BIRTH
Expected redacted: SSN, CREDIT_CARD, MRN, API_KEY, IP_ADDRESS, ORDER_ID
```

5. evaluate_category_toggle.mjs

New LLM eval harness — bridges Layer 1 (LLM quality) and Layer 4 (config tests)

Key capabilities:

- Parses extended metadata from fixtures (Broad category:, Expected redacted:, Expected untouched:, Expected labels:)
- Constructs **redactionConfig** from metadata
- Calls **buildPrompt()** with that config, then runs `lit` inference
- Evaluates 3 dimensions per snippet:
 - **Positive redaction:** enabled-category PII values absent from output
 - **Negative preservation:** disabled-category PII values literally present in output
 - **Wrong-label detection:** output contains a placeholder from a different type than expected (e.g., [PAN 1] where only credit card numbers exist in input)

Outputs JSON report with `positiveRedactions`, `negativePreservations`, `crossContaminations`, `wrongLabels` per snippet.

CLI:

```
node evaluate_category_toggle.mjs [fixture.txt] [--model <path>] [--lit <path>] [--output <json>]
```

Reuses from `evaluate_redaction_snippets.mjs`: `parseSnippetsFile`, `extractStructuredPii`, `runLitInference` (needs `export` added), `STRUCTURED_PII_RULES`.

6. One-line change to evaluate_redaction_snippets.mjs

Export `runLitInference` (line ~244): add `export` keyword so the new harness can reuse it.

Execution Order

Phase 1: Deterministic (no model, <2 seconds)

```
node cross_contamination.test.mjs
node category_isolation.test.mjs
node redaction_category_toggle.test.mjs      # existing
node extension/offscreen_utils.test.mjs     # existing
```

Phase 2: LLM Eval (requires model, 5-15 minutes)

```
node evaluate_category_toggle.mjs test_redaction_category_toggle.txt
```

```
node evaluate_category_toggle.mjs test_cross_contamination.txt
node evaluate_redaction_snippets.mjs test_redaction_snippets.txt  # existing
```

File Summary

Files to Create

File	Type	Purpose
cross_contamination.test.mjs	Deterministic test	Confusable PII pair validation
category_isolation.test.mjs	Deterministic test	"One OFF" isolation for all 7 categories
test_cross_contamination.txt	Fixture	16 snippets targeting confusable pairs
evaluate_category_toggle.mjs	LLM eval harness	Category-aware eval with wrong-label detection

Files to Modify

File	Change
evaluate_redaction_snippets.mjs	Export runLitInference
test_redaction_category_toggle.txt	Add 7 "one OFF" snippets (15-21)

Reference Files (read only)

File	Role
extension/offscreen_utils.mjs	All exported functions for prompt building, validation, category mapping
redaction_category_toggle.test.mjs	makeCategoryConfig pattern and 128-combo test structure
extension/offscreen.js	Retry logic with findDisallowedPlaceholderKeys